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MEMORY SYSTEM WITH DYNAMICALLY ENUMERATED IDENTIFIERS

Abstract

A memory system with high capacity and high performance, having memory units with a type of 3D NOR flash or 3D NAND flash. The memory system comprises a plurality of logic units connected with one another in series to form a chain. Each of the logic units is a logic unit number (LUN) device having an identifier “LUN ID”, and comprises a control circuit. The control circuit communicates with a host controller through a common interface utilizing an ONFI bus and a sideband bus, and the control circuit of a corresponding logic unit may communicate with a control circuit of an adjacent logic unit through the sideband bus. The control circuits of the logic units cooperate with a host controller to perform an enumeration for the identifier “LUN ID” of each of the logic units.

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Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to a semiconductor memory, and more particularly, relates to a memory system with a capability of dynamically enumerating identifiers.

BACKGROUND

[0002] For a memory system, when multiple logic units are integrated in a single package, logic unit number (LUN) addressing mechanism is provided to identify these logic units. Each of the logic units has addressing pads (i.e., LUN addressing pads) used to determine an identifier “LUN ID”.

[0003] The LUN addressing pads, which include “pad LUN1” and “pad LUN0”, are tied to the supply voltage (i.e., VCC) or the ground voltage (i.e., GND) to define the identifier “LUN ID” of the logic unit. Furthermore, a host controller, which is disposed in the host side, may designate a selected logic unit utilizing “bit LUN1” and “bit LUN0” in a fifth address cycle, when address allocation is performed.

[0004] However, as the memory system include a large number of stacked logic units, many LUN addressing pads and bounding wires are needed, which may result in complex masking in fabrication process and lead to a high cost of the memory system.

[0005] To address the above issue, it is desirable to have an improved configuration for memory system, which may dynamically enumerate identifier “LUN ID” for each logic unit.

SUMMARY

[0006] According to an aspect of the disclosure, a memory system is provided. The memory system has high capacity and high performance, which includes a memory unit having a type of NOR or NAND flash (e.g., 3D NAND flash). The memory system comprises a plurality of logic units connected with one another in series to form a chain. Each of the logic units has an identifier, an input data port and an output data port. Each of the logic units comprises a control circuit, which used to control a corresponding one of the logic units to receive a first data signal through the input data port or generate the first data signal internally, and transmit a second data signal through the output data port, and the output data port of the corresponding one of the logic units is coupled to the input data port of an adjacent one of the logic units. Each of the first data signal and the second data signal has a sequence of states, and the identifier of each of the logic units is enumerated according to a count number of a plurality of state transition edges of the states.

[0007] According to another aspect of the disclosure, a memory system is provided. The memory system comprises a plurality of logic units connected with one another in series to form a chain. Each of the logic units has an identifier, an input data port, an output data port and a clock port. Each of the logic units comprises a control circuit, used to control a corresponding one of the logic units to receive a first data signal through the input data port or generate the first data signal internally, transmit a second data signal through the output data port, and receive a clock signal having a sequence of edges through the clock port, and the output data port of the corresponding one of the logic units is coupled to the input data port of an adjacent one of the logic units. The corresponding one of the logic units transmits the second data signal carrying a first command having a start bit, and the adjacent one of the logic units receives the second data signal and latches the start bit of the first command according to the edges of the clock signal, and the identifier of each of the logic units is enumerated according to the start bit of the first command.

[0008] According to still another aspect of the disclosure, a memory system is provided. The memory system comprises a plurality of logic units and a host controller. The logic units are connected with one another in series to form a chain. Each of the logic units has an identifier, a clock port, a plurality of input data ports and a plurality of output data ports, The output data ports of the corresponding one of the logic units are coupled to the input data ports of an adjacent one of the logic units. The host controller is coupled to the input data ports of the first logic unit of the logic units, and coupled to the output data ports of the last logic unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1A is a block diagram of a memory system according to an example of the disclosure.
[0010] FIG. 1B illustrates the memory system of FIG. 1A performing operations with a sideband bus and an ONFI (Open NAND Flash Interface) bus.

[0011] FIG. 1C is a block diagram of another embodiment of the memory system.

[0012] FIG. 1D is a block diagram of another example of the memory system of FIG. 1C.

[0013] FIG. 2 is an exemplary waveform diagram of the data signals, the enable signal and the clock signal of logic units of FIG. 1B.

[0014] FIG. 3A is a block diagram of the memory system illustrating an exemplary enumerating procedure for the identifier “LUN ID” of the logic units.

[0015] FIG. 3B is a block diagram of another embodiment of the memory system.

[0016] FIG. 3C-1 and 3C-2 are block diagrams of still other embodiments of the memory system.

[0017] FIG. 3D-1 and 3D-2 are block diagrams of yet other embodiments of the memory system.

[0018] FIG. 3E is a block diagram of further another embodiment of the memory system.

[0019] FIG. 4 is an exemplary waveform diagram of the data signals of logic units of FIG. 3A.

[0020] FIG. 5 is a block diagram of the memory system illustrating an exemplary enumerating procedure for the identifier “LUN ID” of the logic units.

[0021] FIG. 6 is an exemplary waveform diagram of enable signal, clock signal and data signals of logic units of FIG. 5.

[0022] FIG. 7 is another exemplary waveform diagram of enable signal, clock signal and data signals of logic units of FIG. 5.

[0023] FIG. 8 is a block diagram of memory system according to another example of the disclosure.

[0024] FIG. 9 is an exemplary waveform diagram of enable signal, clock signal, data signals and data signals of logic units of FIG. 8.

[0025] FIG. 10 is a block diagram of memory system according to still another example of the disclosure.

[0026] FIGS. 11A and 11B are exemplary waveform diagrams of enable signal, clock signal, control signal and data signals of logic units of FIG. 10.

[0027] FIGS. 12A and 12B are other exemplary waveform diagrams of enable signal, clock signal, control signal and data signals of logic units of FIG. 10.

[0028] In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically illustrated in order to simplify the drawing.

DETAILED DESCRIPTION

[0029] FIG. 1A is a block diagram of a memory system **1000** according to an example of the disclosure. The memory system **1000** has high capacity and high performance. Referring to FIG. 1A, the memory system **1000** includes a host controller **300** and a memory unit **150**. The host controller **300** may be disposed in the host side, and the host side is, e.g., a central processing unit (CPU) or a micro control unit (MCU). The memory unit **150** includes multiple logic units with a total number of N. In the example of FIG. 1A, the total number N is, e.g., “3”, which refer to three logic units **100**, **101** and **102**. The memory unit **150** may have a type of NOR flash or NAND flash. The memory unit **150** may be a three-dimensional (3D) NOR flash memory unit or a 3D NAND flash memory unit.

[0030] Each of the logic units **100-102** is a logic unit number (LUN) device, which may include a

memory chip or a memory die. Furthermore, each of the logic units **100-102** may include memory arrays with memory blocks therein. The logic units **100-102** are integrated in a package, which is referred to as “multi-chip package (MCP)”. Each of logic units **100-102** has an identifier “logic unit number identity (LUN ID)”. The identifier “LUN ID” is used to provide LUN addressing for identifying the logic units **100-102**. The identifier “LUN ID” of each of the logic units **100-102** is stored in a register, and the identifier “LUN ID” has a registered value.

[0031] Furthermore, each of the logic units **100-102** further includes a control circuit, which serves as a device controller. For example, the logic unit **100** includes a control circuit **10**, the logic unit **101** includes a control circuit **11**, and the logic unit **102** includes a control circuit **12**. The control circuits **10**, **11** and **12** are used to control operations of the logic units **100-102** respectively. The control circuits **10**, **11** and **12** may communicate with the host controller **300** through a common interface utilizing an open-NAND-flash-interface (ONFI) bus and a sideband bus, and the control circuits **10**, **11** and **12** may communicate with one another through the a sideband bus. The control circuits **10**, **11** and **12** cooperate with the host controller **300** to perform an enumeration procedure for the identifier “LUN ID” of each of the logic units **100-102**. In the enumeration procedure, the identifiers “LUN ID” the logic units **100-102** are enumerated to determine the final values of the identifiers “LUN ID”. Such as, the “LUN ID” of the logic unit **100** is enumerated to have a final value “0”. Likewise, the “LUN IDs” of the logic units **101** and **102** are enumerated to have final values “1” and “2”.

[0032] FIG. **1B** illustrates the memory system **1000** of FIG. **1A** performing operations with an ONFI bus **250** and a sideband bus **200**. Referring to FIG. **1B**, the ONFI bus **250** serves as a main memory bus for normal operations of the logic units **100-102**. On the other hand, the sideband bus **200** serves to perform enumeration for the identifiers “LUN ID” of the logic units **100-102**. The sideband bus **200** may be executed with a protocol conformed to the Universal Asynchronous Receiver/Transmitter (UART) or the Universal Serial Bus (USB). The sideband bus **200** utilizes an input data port, an output data port, a clock port and an enable port, and the logic units **100-102** are connected with one another in series to form a chain. In one example, the chain has a type of Daisy Chain. The logic unit **100** has an input data port **SI0**, an output data port **SO0**, a clock port **CLK0** and an enable port **CSB0**. Likewise, the logic unit **101** has an input data port **SI1**, an output data port **SO1**, a clock port **CLK1** and an enable port **CSB1**. The logic unit **102** has an input data port **SI2**, an output data port **SO2**, a clock port **CLK2** and an enable port **CSB2**. Correspondingly, the host controller **300** has an input data port **MISO**, an output data port **MOSI**, a clock port **CLKH** and an enable port **CSBH**.

[0033] In the serial connection forming a chain, the output data port **SO(i)** of the *i*-th logic unit of a current stage is coupled to the input data port **SI(i+1)** of the (i+1)-th logic unit of a next stage. In the example of FIG. **1B**, the output data port **SO0** of the logic unit **100** of a first stage is coupled to the input data port **SI1** of the logic unit **101** of a second stage. The output data port **SO1** of the logic unit **101** of the second stage is coupled to the input data port **SI2** of the logic unit **102** of a last stage. Furthermore, the input data port **SI0** of the logic unit **100** of the first stage is coupled to the output data port **MOSI** of the host controller **300**. Moreover, the output data port **SO2** of the logic unit **102** is coupled to the input data port **MISO** of the host controller **300**. In other words, the logic units **100-102** are connected with one another in series to form the chain, where the logic unit **102** is the last stage of the chain. In the example of FIG. **1B**, the output data port **SO2** of the logic unit **102** (i.e., the last stage) is looped back to the input data port **MISO** of the host controller **300**, and the chain has a form of the Daisy Chain.

[0034] The host controller **300** provides a data signal **da0** through output data port **MOSI**, and data signal **da0** is then received by logic unit **100** through input data port **SI0**. When the control circuit **10** (not shown in FIG. **1B**) controls the logic unit **100** to operate in a non-blocking mode, control circuit **10** controls the data signal **da0** to directly pass to the output data port **SO0**, forming the data signal **da1** (i.e., data signal **da1** is identical with data signal **da0**). When the control circuit **10**

controls the logic unit **100** to operate in a blocking mode, control circuit **10** controls the data signal **da0** to be delayed by a period, and then pass to the output data port **SO0** (i.e., data signal **da1** is a delayed-version of data signal **da0**). In addition, control circuit **10** may control to generate data signal **da1** regardless data signal **da0** (i.e., data signal **da1** is not related to data signal **da0**). With respect to the logic unit **100** of current stage, the data signal **da0** is referred as the “first data signal”, and the data signal **da1** is referred as the “second data signal”. That is, the logic unit **100** receives the “first data signal” through the input data port **SI0** and transmits the “second data signal” through the output data port **SO0**.

[0035] Data signal **da1** is received by logic unit **101** through input data port **SI1** and then provided to output data port **SO1**, forming data signal **da2**. Then, data signal **da2** is received by logic unit **102** through input data port **SI2** and then provided to output data port **SO2**, forming data signal **da3**. Thereafter, data signal **da3** is sent back to host controller **300** through input data port **MISO**.

[0036] The host controller **300** may analyze the data signal **da3** provided by the output data port **SO2** of the logic unit **102**, and the host controller **300** may know the connecting manner of the logic units **100-102**. In one example, the output data port **SO2** of the logic unit **102** may transmit the data signal **da3** with a voltage **V0** (e.g., the voltage **V0** has a high voltage level of 1.5V, 3V or 5V, etc.) to indicate the host controller **300** that, the logic unit **102** is the last stage of the chain.

[0037] Alternatively, referring to FIG. 1C which illustrates another embodiment of the memory system **1001** (in FIG. 1C, only the sideband bus **200** is shown, and the ONFI bus **250** is omitted), the output data port **SO2** of the logic unit **102** may not be looped back to the host controller **300**. Instead, the output data port **SO2** of the logic unit **102** is tied to a voltage **V1** to indicate that the logic unit **102** is the last stage of the chain. In the embodiment of FIG. 1C, the output data port **SO2** of the logic unit **102** is tied to a ground end **GND**, hence the voltage **V1** for the output data port **SO2** is a ground voltage (e.g., 0V). In another embodiment shown in FIG. 1D, the output data port **SO2** of the logic unit **102** is tied to a voltage **V1** with a high voltage level of 1.2V, 1.8V or 3V, etc.

[0038] Referring to FIG. 1B again, a clock signal **ck** is generated by the host controller **300** through the clock port **CLKH**, and then provided to the logic units **100-102** through their respective clock ports **CLK0-CLK2**. A content **d(0)** is carried by data signal **da0**, and control circuit **10** controls logic unit **100** to latch data signal **da0** to acquire content **d(0)** according to rising edges of the clock signal **ck**. Then, control circuit **10** controls logic unit **100** to transmit data signal **da1** with content **d(0)** according to falling edges of the clock signal **ck**. Similarly, at a rising edge of a next clock cycle of clock signal **ck**, control circuit **10** controls to latch data signal **da0** to acquire a next content **d(1)**. Then, the content **d(1)** will be carried by data signal **da0** and transmitted through output data port **SO0** at a falling edge of the clock signal **ck**. In this manner, contents **d(0)**, **d(1)**, . . . etc. may be sequentially provided to logic units **100-102**. In one example, contents **d(0)**, **d(1)**, . . . etc. may be a sequence of state(s). In another example, contents **d(0)**, **d(1)**, . . . etc. may be a series of data, such as commands of SPI (Serial Peripheral Interface) protocol.

[0039] In addition, enable signal **en** may be generated by the host controller **300** through the enable port **CSBH**, and then be provided to the logic units **100-102** through their respective enable ports **CSB0-CSB2**. These enable ports **CSB0-CSB2** are low-active. Initially, the enable signal **en** has a high voltage level (e.g., 1.2V, 1.8V or 3V), and the enable ports **CSB0-CSB2** are de-asserted and the logic units **100-102** are de-activated. When the enable signal **en** changes to a low voltage level (e.g., 0V), the enable ports **CSB0-CSB2** are asserted and the logic units **100-102** are activated to operate with the sideband bus **200**. After the sideband bus **200** is activated, an enumerating procedure for the identifier “LUN ID” of logic units **100-102** may start to perform.

[0040] FIG. 2 is an exemplary waveform diagram of the data signals **da0-da2**, the enable signal **en** and the clock signal **ck** of logic units **100** and **101** of FIG. 1B, illustrating the transmission of data signals **da0-da2**. Referring to FIG. 2, initially the enable signal **en** has the high voltage level, and enable port **CSB0** and **CSB1** of logic unit **100** and **101** are de-asserted and logic units **100** and **101** are not activated. Input data ports **SI0** and **SI1** of logic unit **100** and **101** are not driven and have

high impedance, and output data ports SO0 and SO1 each have a high voltage level.

[0041] When enable signal en changes to the low voltage level, enable port CSB0 and CSB1 of logic unit **100** and **101** are asserted and logic units **100** and **101** are activated. Input data port SI0 is provided with data signal da0 which carries a content d(0). At a rising edge r1 of clock cycle ck(1) of clock signal ck, control circuit **10** of logic unit **100** controls to latch data signal da0 to acquire content d(0). Then, at a falling edge f1 of the same clock cycle ck(1), control circuit **10** controls to drive output data port SO0 to transmit data signal da1 such that data signal da1 carries content d(0).

[0042] The data signal da1 is provided to input data port SI1 of logic unit **101**, and control circuit **11** of logic unit **101** controls to latch data signal da1 to acquire content d(0) at rising edge r2 of a next clock cycle ck(2). Then, at a falling edge f2 of clock cycle ck(2), control circuit **11** controls to drive output data port SO1 to transmit data signal da2 which carries content d(0).

[0043] On the other hand, at rising edge r2 of clock cycle ck(2), data signal da0 is latched at input data port SI0 to acquire a next content d(1). Then, at falling edge f2 of clock cycle ck(2), data signal da1 is driven at output data port SO0 to carry content d(1).

[0044] After sideband bus **200** operates for several clock cycles, e.g., three clock cycles ck(1), ck(2) and ck(3), if enable signal en changes to high voltage level again, enable port CSB0 and CSB1 of logic unit **100** and **101** are de-asserted again, and logic units **100** and **101** are de-activated. Output data ports SO0 and SO1 each change to high voltage level again.

[0045] FIG. 3A is a block diagram of the memory system **1000** illustrating an exemplary enumerating procedure for the identifier “LUN ID” of the logic units **100-102**. In the exemplary enumerating procedure of FIG. 3A, the memory system **1000** may utilize input data ports SI0-SI2 and output data ports SO0-SO2 of logic units **100-102** in the sideband bus **200** to enumerate identifier “LUN ID”, but not utilize the clock ports CLK0-CLK2 and the enable ports CSB0-CSB2. Furthermore, the host controller **300** may be selectively utilized for the enumerating procedure. When the host controller **300** is utilized, the output data port MOSI of the host controller **300** is coupled to the input data port SI0 of logic unit **100**, and host controller **300** generates data signal da0 and provides data signal da0 to logic unit **100**. The input data port MISO of host controller **300** is coupled to the output data port SO2 of logic unit **102**, and host controller **300** receives data signal da3 from logic unit **102**.

[0046] The control circuits **10**, **11** and **12** illustrated in FIG. 1A (not shown in FIG. 3A) control the logic units **100**, **101** and **102** to perform enumerating procedure for identifier “LUN ID” with sideband bus **200**. In the example of FIG. 3A, control circuits **10**, **11** and **12** control to transmit and receive data signals da0-da3 through input data ports SI0-SI2 and output data ports SO0-SO2 of sideband bus with SPI protocol. The identifier “LUN ID” of logic units **100**, **101** and **102** are enumerated according to data signals da0-da3. In an example, final values of the enumerated identifiers “LUN ID” of logic units **100-102** may be provided to host controller **300** through data signal da3. In another example, final values of the enumerated identifiers “LUN ID” may be provided to another control circuit (not shown in FIG. 3A) in the memory system **1000**, other than control circuits **10-12** respective disposed in logic units **100-102**. In still another example, final values of the enumerated identifiers “LUN ID” may be provided to a register or a storage element (not shown in FIG. 3A) in the memory system **1000**.

[0047] In another embodiment of memory system **1002** shown in FIG. 3B, the input data port SI0 of logic unit **100** may receive the data signal da0 from the host controller **300**. Furthermore, the output data port SO2 of logic unit **102** is looped back to couple to the input data port MISO of host controller **300**. In other examples, the logic unit **100** itself may generate the data signal da0 with an internal on-die weak connection (e.g., an internal resistor R1).

[0048] More particularly, taking the memory system **1003** of FIG. 3C-1 as an example, the resistor R1 may be a pull-up resistor which may be disposed around the input data port SI0. When the memory system **1003** is powered-ON or is reset, the resistor R1 may generate the data signal da0

with a high voltage level and results in a rising edge r1 of the data signal da0). Alternatively, in another example shown in FIG. 3C-2, the resistor R1 may be a pull-down resistor which generates a data signal da0 with a low voltage level (e.g., ground voltage).

[0049] In yet another embodiment of memory system 1004 shown in FIG. 3D-1, the input data port SIO of the logic unit 100 may be externally tied to a voltage V1 with a ground voltage, indicating that the logic unit 100 is the first stage of the chain. Alternatively, in another example of FIG. 3D-2, the input data port SIO of the logic unit 100 may be externally tied to a voltage V1 with a high voltage level (e.g., 1.2V, 1.8V or 3V, etc.).

[0050] In further another embodiment of memory system 1005 shown in FIG. 3E, the memory system 1005 may further include an aux controller 400 other than the host controller 300, and the logic units 100-102 may cooperate with the aux controller 400. The aux controller 400 is, e.g., a buffer chip, which serves to share the work load of the host controller 300. The input data port SIO of the logic unit 100 (i.e., the first stage of the chain) may be coupled to the aux controller 400, and data signal da0 may be generated by the aux controller 400 (instead of the host controller 300). Furthermore, the output data port SO2 of the logic unit 102 (i.e., the last stage of the chain) may be coupled to the aux controller 400 to convey the data signal, and aux controller 400 may be configured to determine the total number of the logic units 100-102 according to the data signal da3. The host controller 300 access the logic units 100-102 through the aux controller 400 with the regular ONFI bus along or in tandem with the help of the sideband bus.

[0051] FIG. 4 is an exemplary waveform diagram of the data signals da0-da3 of logic units 100-102 of FIG. 3A, illustrating the enumerating procedure for the identifiers “LUN ID” of logic units 100-102. Referring to both FIGS. 3A and 4, when memory system 1000 enters the enumerating procedure for identifier “LUN ID”, the host controller 300 is configured to transmit data signal da0, and control circuits 10-12 (not shown in FIG. 3A) respectively control logic units 100-102 to transmit data signals da1-da3. The data signals da0-da3 carry predefined patterns to facilitate enumerating identifier “LUN ID”. In the example of FIG. 4, data signal da0 has a “pattern of a constant level, which has a high voltage level. The host controller 300 drives the data signal da0 through the output data port MOSI, such that the data signal da0 changes from the low voltage level to the high voltage level at a rising edge r1.

[0052] Each of data signals da1-da3 has a pattern of sequence of state(s). The identifiers “LUN ID” are enumerated according to the count number of state transition edges in data signals da1-da3. The edges of the data signals da1-da3 may refer to rising edges or falling edges for various of examples. The control circuits 10-12 of the logic units 100-102 may have a detector (not shown in the figure), and the detector may be configured to detect the rising edges or falling edges of the data signals da1-da3.

[0053] Then, taking the memory system 1000 of FIG. 3A as an example, in which the data signal da0 is provided by the host controller 300. The host controller 300 pulls up the data signal da0 from low voltage level to high voltage level, then the data signal da0 keeps at a constant level. Logic unit 100 receives data signal da0 at input data port SIO, and control circuit 10 in logic unit 100 may detect a rising edge r1 of data signal da0. Rising edge r1 may be referred as a “first rising edge” of data signal da0 during the enumerating procedure. The “first rising edge” of the pulses of the data signal da0 is one of the rising edges of the pulses which is firstly detected by control circuit 10 when entering the enumerating procedure.

[0054] In response to the detection of rising edge r1, the registered value of identifier “LUN ID” of logic unit 100 obtains an initial value. The initial value is set by the control circuit 10, such as, the initial value is set as “0”. Then, control circuit 10 continues to detect further rising edges of data signal da0. A first period ET starting from the rising edge r1 is defined, for example, first period ET is longer than two times of a pulse duration PD of a pulse in data signals da1-da3. When the first period ET expires and control circuit 10 does not detect any further rising edges (other than rising edge r1), the registered value of identifier “LUN ID” of logic unit 100 has a final value determined

as “0”.

[0055] Then, control circuit **10** controls logic unit **100** to generate data signal da1 with a sequence of pulses, and these pulses have a count number M. In the example of FIG. 4, the count number M of the pulses is equal to the final value of identifier “LUN ID” of logic unit **100** added by “2”, as shown in equation (1):

$$[00001] \ M = \text{finalvalueofLUN_ID} + 2 \quad (1)$$

[0056] The “LUN ID” of logic unit **100** has a final value determined as “0”, hence the count number M is equal to “2”. Therefore, control circuit **10** controls logic unit **100** to generate data signal da1 with two pulses p1(1) and p1(2). These pulses p1(1) and p1(2) are transmitted to logic unit **101** through input data port SI1.

[0057] Then, control circuit **11** in logic unit **101** is configured to detect rising edges of data signal da1. When control circuit **11** detects a rising edge r1 of data signal da1, a registered value of identifier “LUN ID” of logic unit **101** obtains an initial value (e.g., “0”), which is set by the control circuit **11**. The rising edge r1 of data signal da1 may be referred as a “first rising edge” of data signal da1 during the enumerating procedure. When entering the enumerating procedure, data signal da1 at input data port SI1 is changed from the low voltage level to the high voltage level at the rising edge r1. The rising edge r1 of data signal da1 may be referred as a “first rising edge”. After the rising edge r1, control circuit **11** detects further rising edges of data signal da1. When a subsequent rising edge r2 after rising edge r1 is detected by control circuit **11**, registered value of “LUN ID” of logic unit **101** is incremented by “1” (i.e., incremented from “0” to become “1”). The rising edge r2 may be referred to as a “second rising edge” of data signal da1, with respect to the first rising edge r1.

[0058] Still, control circuit **11** continues to detect further rising edges. When a first period ET (starting from the rising edge r2) expires and control circuit **11** does not detect any further rising edges (other than rising edges r1 and r2), the registered value of identifier “LUN ID” of logic unit **101** has a final value determined as “1”. In other words, rising edge r2 may also be referred to as “last rising edge” detected by control circuit **11**, the “last rising edge” of the pulses of the data signal da1 is one of the rising edges of the pulses which is last detected by control circuit **11**. The registered value of identifier “LUN ID” of logic unit **101** has a determined final value, when first period ET starting from the last rising edge r2 expires (i.e., the final value of the identifier “LUN ID” is determined when first period ET expires).

[0059] Then, control circuit **11** controls logic unit **101** to generate data signal da2 as a sequence of M pulses, with the count number M equal to the final value of identifier “LUN ID” of logic unit **101** added by “2”. In the example of FIG. 4, final value of identifier “LUN ID” of logic unit **101** is “1”, hence logic unit **101** to generate data signal da2 as three pulses p2(1), p2(2) and p2(3).

[0060] Similarly, the pulses p2(1), p2(2) and p2(3) are received by logic unit **102** of next stage at its input data port SI2, and control circuit **12** of logic unit **102** detects rising edges r1-r3 of pulses p2(1)-p2(3) to enumerate “LUN ID” of logic unit **102**. When the rising edge r1 (i.e., the first rising edge) is detected, the registered value of “LUN ID” of logic unit **102** obtains an initial value (e.g., “0”). Then, when rising edge r2 (i.e., the second rising edge) and rising edge r3 (i.e., the third rising edge) are detected, the registered value of “LUN ID” of logic unit **102** is incremented by “1” for each of rising edges r2 and r3. That is, registered value of “LUN ID” of logic unit **102** becomes “2”. Then, if first period ET (starting from the rising edge r3) expires and control circuit **12** does not detect any further rising edges (other than rising edges r1-r3), the registered value of identifier “LUN ID” of logic unit **102** has a final value determined as “2”. In other words, rising edge r3 may also be referred to as “last rising edge” detected by control circuit **12**, and registered value of identifier “LUN ID” of logic unit **102** has a final value of “2”, which is determined when first period ET starting from the last rising edge r3 expires. Then, control circuit **12** controls logic unit **102** to generate data signal da3 as four pulses p3(1)-p3(4), with the count number M equal to “4”.

[0061] In the example of FIGS. 3 and 4, when control circuit of the i-th logic unit detects subsequent rising edges r2, r3, r4 . . . etc. after the first rising edge r1, registered value of “LUN ID” is incremented by “1”. Furthermore, the i-th logic unit generates M pulses (where M equal to final value plus “2”) to (i+1)-th logic unit of next stage. Therefore, final value of “LUN ID” of the i-th logic unit is determined as “(i-1)”, as equation (2). Such as, the first logic unit (i.e., logic unit **100**) has “LUN ID” with final value of “0”, the second logic unit (i.e., logic unit **101**) has “LUN ID” with final value of “1”, and the third logic unit (i.e., logic unit **102**) has “LUN ID” with final value of “2”.

$$[00002] \text{finalvalueofLUNID} = (i - 1) \quad (2)$$

[0062] In another example, when control circuit of the i-th logic unit detects subsequent rising edges after the first rising edge, registered value of “LUN ID” may increase by an increment “K1” (where K1 is an positive integer such as 1, 2, 3, 4 . . . etc.) Correspondingly, the i-th logic unit may generate M pulses with M equal to final value plus “(2×K1)”. In this manner, final value of “LUN ID” of the i-th logic unit may be determined as “(i×K1-1)”, as equation (3). Such as, when K1 is equal to “2”, the first logic unit (i.e., logic unit **100**) has “LUN ID” with final value of “1”, the second logic unit (i.e., logic unit **101**) has “LUN ID” with final value of “3”, and the third logic unit (i.e., logic unit **102**) has “LUN ID” with final value of “5”.

$$[00003] \text{finalvalueofLUNID} = (i \times K1 - 1) \quad (3)$$

[0063] The pulses generated by the N-th logic unit (i.e., the logic unit of the last stage in the chain) may be received by the host controller **300** through the input data port MISO. The host controller **300** may determine the total number N of the logic units (for example, total number “3” of the logic units **100-102**) according to the pulses generated by the N-th logic unit. In one example, the host controller **300** may determine the total number N of the logic units according to the count number M of the pulses generated by the N-th logic unit. Such as, the host controller **300** receives four pulses p3(1)-p3(4) from the N-th logic unit, and the host controller **300** may analyze the count number M (i.e., equal to “4”) of the four pulses p3(1)-p3(4), and then determine the total number N of the logic units as the count number M minus one (i.e., N=M-1).

[0064] In the above example shown in FIG. 4, the identifier “LUN ID” of the i-th logic unit is enumerated according to the rising edges of data signal da0-da3. In other examples (not shown), the identifier “LUN ID” may be enumerated according to the falling edges of data signal da0-da3. For example, in response to the detection of fall edge f1 (not shown) of data signal da1, a registered value of identifier “LUN ID” of logic unit **101** obtains an initial value of “0”. Then, when a subsequent falling edge f2 after the falling edge f1 is detected by control circuit **11**, registered value of “LUN ID” of logic unit **101** is incremented by “1”, such that registered value becomes “1”. Then, when the first period ET after the falling edge f2 expires and control circuit **11** does not detect any further falling edges (other than falling edge f2), the registered value of identifier “LUN ID” of logic unit **101** has a determined final value. Similar schemes may be utilized by the logic unit **102**. Such as, when the control circuit **12** of logic unit **102** detects fall edge f1 of data signal da2, a registered value of identifier “LUN ID” of logic unit **102** obtains an initial value of “0”. When the control circuit **12** detects a next fall edge f2 of data signal da2, registered value of identifier “LUN ID” of logic unit **102** is incremented by “1” to form “1”, and registered value of identifier “LUN ID” of logic unit **102** is further incremented as “2” when a next falling edge f3 is detected. Then, when the first period ET after the falling edge f3 expires and no any further falling edges are detected, the registered value of identifier “LUN ID” of logic unit **102** has a determined final value.

[0065] FIG. 5 is a block diagram of the memory system **1000** illustrating an exemplary enumerating procedure for the identifier “LUN ID” of the logic units **100** and **101**. FIG. 5 shows two logic units **100** and **101** as an example, but the total number of logic units is not limited as “2”.

[0066] Unlike the example of FIG. 3A, the exemplary memory system **1000** of FIG. 5 may utilize

input data ports SIO and SI1, output data ports SO0 and SO1, clock ports CLK0 and CLK2 and enable ports CSB0 and CSB1 of logic units **100** and **101** in the sideband bus **200**. Furthermore, the host controller **300** is utilized, with its output data port MOSI coupled to the input data port SIO of logic unit **100** and its input data port MISO coupled to the output data port SO2 of logic unit **102**. Host controller **300** provides data signal da0 to logic unit **100** and receives data signal da2 from logic unit **101**. Furthermore, host controller **300** provides enable signal en at enable port CSBH and transmits enable signal en to logic units **100** and **101**. Moreover, host controller **300** provides clock signal ck at clock port CLKH and transmits clock signal ck to logic units **100** and **101**.

[0067] Furthermore, memory system **1000** may further include another control circuit (which is not shown in FIG. 5) other than control circuits **10-12** as shown in FIG. 1A respectively disposed in logic units **100** and **101**. Such a control circuit may record final values of the enumerated identifiers “LUN ID” of logic units **100** and **101**. In another example, the host controller **300** may further include a register or a storage element (not shown in FIG. 5) to record final values of the enumerated identifiers “LUN ID” of logic units **100-102**.

[0068] FIG. 6 is an exemplary waveform diagram of enable signal en, clock signal ck and data signals da0-da2 of logic units **100** and **101** of FIG. 5, illustrating another enumerating procedure for the identifiers “LUN ID” of logic units **100** and **101**. Referring to both FIGS. 5 and 6, before the enumerating procedure for the identifier “LUN ID”, during or after a power-on cycle of the memory system **1000** in which the host controller **300** and logic units **100** and **101** are powered-ON, or, during or after a reset cycle of the memory system **1000** for which the host controller **300** and logic units **100** and **101** are reset, the registered value of “LUN ID” of each of logic units **100** and **101** obtains an initial value (e.g., “0”). Furthermore, in an idle state after the power-on cycle or the reset cycle, the output data ports SO0 and SO1 are initially driven as a voltage V2. The voltage V2 may represent the idle state, and the voltage V2 may indicate a logic value opposite to the logic value of the “start bit” of the first command carried by the data signal da0, as will be discussed in later paragraphs.

[0069] When entering the enumerating procedure for identifier “LUN ID”, enable signal en changes to the low voltage level, and enable ports CSB0 and CSB1 of logic units **100** and **101** are asserted, hence logic units **100** and **101** are activated to operate the sideband bus **200**. After enable signal en changes to the low voltage level, clock signal ck starts to toggle with a sequence of clock cycles with respective rising edges and falling edges. Such as, clock signal ck has a rising edge r1 and a falling edge f1 in the first clock cycle. Rising edge r1 may be referred as a “first rising edge” of clock signal ck during the enumerating procedure, such as, a “first rising edge” after enable signal en changes to the low voltage level (i.e., after enable ports CSB0 and CSB1 are asserted and logic units **100** and **101** are activated to operate the sideband bus **200**). Falling edge f1 may be referred as a “first falling edge” of clock signal ck, after enable signal en changes to the low voltage level. Furthermore, clock signal ck has subsequent rising edges r2, . . . r8, r9 and r10 after the first rising edge r1.

[0070] In the beginning of the enumerating procedure for identifier “LUN ID”, after enable signal en changes to the low voltage level, host controller **300** transmits data signal da0 to logic unit **100** at input data port SIO, and data signal da0 carries a first command of SPI protocol. The first command has totally 8 bits including 1 bit of “start bit” and 7 bits of op-code, and the last bit of the op-code refers to an “end bit”. The “start bit” of the first command has a logic value which may be “1” (i.e., corresponding to high voltage level) or “C” (i.e., corresponding to low voltage level), and the logic value of the “start bit” may be opposite to the logic value of an “idle state” of the sideband bus. When the “idle state” has a logic value of “1” (i.e., the “idle state” is represented by logic “1”), the “start bit” has a logic value of “C”. On the other hand, when the “idle state” has a logic value of “C”, the “start bit” has a logic value of “1”. In the example of FIG. 6, the first command may be an “Enumeration Command” with binary bits of “01110100” representing a hex number of “0x74”, with the start bit of “C” and the end bit of “C”.

[0071] In the idle state, the output data ports SO0 and SO1 are initially driven as the voltage V2, and the voltage V2 represents a logic value opposite to the logic value of the “start bit” of the first command. In the example of FIG. 6, the “start bit” of the first command has a logic value “C”, and the voltage V2 has a high voltage level representing a logic value “1” opposite to the “start bit”.

[0072] The control circuit 10 (not shown in FIG. 5) of logic unit 100 detects rising edges and falling edges of clock signal ck. When control circuit 10 detects the first rising edge r1, control circuit 10 controls logic unit 100 to latch data signal da0 at input data port SI0, and the start bit of the first command is acquired by logic unit 100. Since the start bit is acquired, identifier “LUN ID” of logic unit 100 keeps its registered value unchanged (i.e., keeps the registered value as the initial value “C” which is set during the power-on cycle or the reset cycle of memory system 1000).

[0073] On the other hand, when control circuit 11 (not shown in FIG. 5) of logic unit 101 of next stage detects the first rising edge r1 of clock signal ck, the control circuit 11 of logic unit 101 latches data signal da1 at first rising edge r1 and acquire voltage V2 in data signal da1. That is, at the first rising edge r1, the data signal da1 is latched to acquire voltage V2 indicating the idle state. Therefore, the logic unit 101 knows that, at the first rising edge the data signal sa1 remains at the idle state, and the start bit of the first command has not been acquired. Hence, identifier “LUN ID” of logic unit 101 increments its registered value by “1”. In another example, registered value of the identifier “LUN ID” may be incremented by an increment “K2” (where K2 is a positive integer of 2, 3, 4 . . . etc.).

[0074] After the rising edge r1, control circuit 10 of logic unit 100 detects a falling edge f1 of clock signal ck, and control circuit 10 controls the logic unit 100 to pass data signal da0 to output data port SO0, forming data signal da1. Data signal da1 is then provided to input data port SI1 of logic unit 101.

[0075] After the falling edge f1, control circuit 10 of logic unit 100 detects a rising edge r2 (i.e., the second rising edge) of clock signal ck, and logic unit 100 latches data signal da0 and acquires a second bit “1” of the first command. Furthermore, control circuit 11 of logic unit 101 detects the rising edge r2 of clock signal ck, and logic unit 101 latches data signal da1 and acquires the start bit of the first command (i.e., the start bit has a logic value opposite to that represented by the second voltage V2). Since the start bit has been acquired by logic unit 101 (not the second voltage V2 of the idle state is acquired), identifier “LUN ID” of logic unit 101 keeps its registered value unchanged (i.e., keeps its registered value as “1”).

[0076] Then, control circuit 10 of logic unit 100 detects a rising edge r3 (i.e., the third rising edge) of clock signal ck, and logic unit 100 latches data signal da0 and acquires a third bit “1” of the first command. Furthermore, control circuit 11 of logic unit 101 detects the rising edge r3 of clock signal ck, and logic unit 101 latches data signal da1 and acquires the second bit “1” of the first command.

[0077] Likewise, control circuit 10 of logic unit 100 detects subsequent rising edges r4, r5, r6 and r7 (i.e., the 4-th to the 7-th rising edges) of clock signal ck, and logic unit 100 latches data signal da0 at these rising edges r4, r5, r6 and r7 respectively and acquires a 4-th bit “1”, a 5-th bit “0”, a 6-th bit “1” and a 7-th bit “0” of the first command.

[0078] Then, at a rising edge r8, which is subsequent to the rising edges r4, r5, r6 and r7, logic unit 100 latches data signal da0 to acquire the end bit “0” (i.e., the 8-th bit) of the first command. The end bit of the first command is acquired, and the control circuit 10 of the logic unit 100 checks whether the totally acquired 8 bits, which are latched at rising edges r1-r8 respectively, conform to a predefined command content of the SPI protocol of the sideband bus. When the control circuit 10 of the logic unit 100 verifies that the acquired 8 bits (i.e., the start bit to the end bit) of the first command exactly meet a content “01110100” of the “Enumeration Command”, the registered value of the identifier “LUN ID” of logic unit 100 is determined as a final value of “0”. On the other hand, when the acquired 8 bits of the first command are verified not to meet the content “01110100” of the “Enumeration Command”, the registered value of the identifier “LUN ID” of

logic unit **100** is kept unchanged.

[0079] Similar schemes may be applied to the logic unit **101**. At a rising edge **r9** subsequent to the rising edge **r8**, logic unit **101** latches data signal **da1** to acquire the end bit of the first command. When the control circuit **11** of the logic unit **101** verifies that the acquired 8 bits of the first command meet a content “01110100” of “Enumeration Command”, the registered value of the identifier “LUN ID” of logic unit **101** is determined as a final value of “1”. Otherwise, if not to meet the content “01110100” of the “Enumeration Command”, the registered value of the identifier “LUN ID” of logic unit **101** is kept unchanged.

[0080] On the other hand, when a falling edge **f2** of clock signal **ck** is detected by control circuit **11** of logic unit **101**, control circuit **11** control logic unit **101** to pass data signal **da1** to output data port **SO1**, forming data signal **da2**. Data signal **da2** is then provided to input data port **MISO** of host controller **300**. Then, at a rising edge **r10** subsequent to the rising edge **r9**, host controller **300** latches data signal **da2** to acquire the end bit of the first command, and host controller **300** controls to finish the enumerating procedure for identifiers “LUN ID” of logic units **100** and **101**. Then, the enable signal **en** changes to high voltage level, and logic units **100** and **101** are de-activated.

[0081] FIG. 7 is another exemplary waveform diagram of enable signal **en**, clock signal **ck** and data signals **da0-da2** of logic units **100** and **101** of FIG. 5, illustrating a status-reporting procedure for logic units **100** and **101**. The status-reporting procedure of logic units **100** and **101** may be performed after the enumerating procedure for their identifiers “LUN ID”. In the example of FIG. 7, data signals **da0-da2** may carry a second command to facilitate the status-reporting procedure. The second command is, for example, a “Report Command”. The “start bit” of the second command has a predefined logic value **L_S** which is similar to the “start bit” of the first command. The predefined logic value **L_S** of the “start bit” of the second command may be opposite to the logic value representing an “idle state” of the sideband bus. On the other hand, the “end bit” of the second command has a predefined logic value **L_E** may be opposite to predefined logic value **L_S** of the “start bit”. In the example of FIG. 7, the second command (e.g., the “Report Command”) has binary bits of “00000101” (i.e., representing a hex number of “0×05”) with a start bit of “0” and an end bit of “1”.

[0082] Before the status-reporting procedure, enable signal **en** may return to high voltage level. When entering the status-reporting procedure, enable signal **en** changes to low voltage level, and clock signal **ck** starts to toggle. At a rising edge **r1'** (i.e., the “first rising edge” after enable signal **en** changes to low voltage level), control circuit **10** of logic unit **100** starts to receive the Report Command carried by data signal **da0**. Then, at a falling edge **f1'**, logic unit **100** drives data signal **da0** to output data port **SO0**, forming data signal **da1**. Logic unit **101** of next stage may start to receive Report Command carried by data signal **da1**.

[0083] Then, at a rising edge **r8'** (which is among several subsequent rising edges after the rising edge **r2'**), logic unit **100** latches data signal **da0** to acquire the end bit of Report Command. Therefore, logic unit **100** is ready to output a status data **SR0** which indicates a status of logic unit **100**. The status data **SR0** may have a data length of **L** bits. Then, at a subsequent falling edge **f9'** of a next clock cycle with respect to rising edge **r8**, logic unit **100** starts to transmit the **L**-bit status data **SR0** in data signal **da1** through its output data port **SO0**, regardless data signal **da0** at its input data port **SI0**. Clock cycles of a number **L** are needed to transmit the **L**-bit status data **SR0** in data signal **da1**.

[0084] The status data **SR0** carried by data signal **da1** is received by logic unit **101**. At a subsequent falling edge **f10'**, the status data **SR0** in data signal **da1** passes to output data port **SO1** of logic unit **101**, and control circuit **11** controls logic unit **101** to wait **L** clock cycles (starting from falling edge **f10'**) to complete transmission of status data **SR0**. Then, at a falling edge **f(10+L)'** which lags the falling edge **f10'** by **L** clock cycles, logic unit **101** starts to transmit its status data **SR1** through its output data port **SO1**, regardless the data signal **da1** at its input data port **SI1**.

[0085] In the above example of status-reporting procedure, logic unit **101** has identifier “LUN ID”

of “1”, and logic unit **101** may wait (L+2)'s clock cycles for completing transmission of status data SR0, and then transmit its status data SR1. When memory system **1000** includes more logic units, such as logic units **102**, **103**, . . . etc., logic unit **102** with identifier “LUN ID” of “2” may wait (L+3)'s clock cycles before transmitting its status data SR2, and logic unit **103** with identifier “LUN ID” of “3” may wait (L+4)'s clock cycles before transmitting its status data SR3. That is, the (i+1)-th logic unit with identifier “LUN ID” of “i” may wait (i+L+1) clock cycles before transmitting its status data SR(i).

[0086] On the other hand, at the first rising edge (i.e., rising edge r1') and the second rising edge (i.e., rising edge r2') after enable signal en changes to low voltage level, host controller **300** latches data signal da2 having voltage VH at its data port MISO. Since host controller **300** latches voltage VH at two rising edges r1' and r2', host controller **300** may recognize that logic units of total number “2” (i.e., two logic units **100** and **101**) exist in memory system **1000**.

[0087] FIG. **8** is a block diagram of memory system **1000b** according to another example of the disclosure. The memory system **1000b** of FIG. **8** is similar to the memory system **1000** of FIG. **5** except that, each of logic units **100** and **101** has two data paths operating with two input data ports and two output data ports. More particularly, the logic unit **100** has two input data ports SI0a and SI0b and two output data ports SO0a and SO0b. The logic unit **101** has two input data ports SI1a and SI1b and two output data ports SO1a and SO1b. Furthermore, host controller **300** has two input data ports MISOa and MISOb and two output data ports MOSIa and MOSIb.

[0088] To operate one of the two data paths of memory system **1000b**, host controller **300** transmits data signal da0(a) through output data port MOSIa, data signal da0(a) is provided to logic unit **100** at input data port SI0a. Then, logic unit **100** transmits data signal da1(a) through output data port SO0a, data signal da1(a) is provided to logic unit **101** at input data port SI1a. Then, logic unit **101** transmits data signal da2(a) through output data port SO1a, data signal da2(a) is provided to host controller **300** at input data port MISOa.

[0089] To operate the other one of the two data paths of memory system **1000b**, host controller **300** transmits data signal da0(b) through output data port MOSIb, data signal da0(b) is provided to logic unit **100** at input data port SI0b. Then, logic unit **100** transmits data signal da1(b) through output data port SO0b, data signal da1(b) is provided to logic unit **101** at input data port SI1b. Then, logic unit **101** transmits data signal da2(b) through output data port SO1b, data signal da2(b) is provided to host controller **300** at input data port MISOb.

[0090] FIG. **9** is an exemplary waveform diagram of enable signal en, clock signal ck, data signals da0(a)-da2(a) and data signals da0(b)-da2(b) of logic units **100** and **101** of FIG. **8**. Referring to FIG. **8**, when enable signal en changes to low voltage level, enable port CSB0 and CSB1 of logic unit **100** and **101** are asserted and logic units **100** and **101** are activated. Input data ports SI0a and SI0b of logic unit **100** are provided with data signals da0(a) and da0(b) which carry contents D7 and D6.

[0091] At a rising edge r1 of clock cycle ck(1) of clock signal ck, control circuit **10** (not shown in FIG. **8**) of logic unit **100** controls to latch data signal da0(a) and da0(b) at input data ports SI0(a) and SI0(b), acquiring contents D7 and D6. Then, at a falling edge f1 of the same clock cycle ck(1), control circuit **10** controls to transmit data signals da1(a) and da1(b) which carry contents D7 and D6, to be provided to input data ports SI1(a) and SI1(b) of logic unit **101**.

[0092] Similarly, at a rising edge r2 of clock cycle ck(2) of clock signal ck, control circuit **11** (not shown in FIG. **8**) of logic unit **101** controls to latch data signal da1(a) and da1(b) at input data ports SI1(a) and SI1(b), acquiring contents D7 and D6. Then, at a falling edge f2 of the same clock cycle ck(2), control circuit **11** controls to transmit data signals da2(a) and da2(b) which carry contents D7 and D6, to be provided to input data ports MISO(a) and MISO(b) of host controller **300**.

[0093] In the examples of FIGS. **8** and **9**, the i-th logic unit has two data paths operating with two input data ports SI(i-1)(a) and SI(i-1)(b) and two output data ports SO(i-1)(a) and SO(i-1)(b).

Therefore, transmission bandwidth are enhanced by two times, hence processing time of enumerating procedure for “LUN ID” may be reduced. Such as, only half the clock cycles are needed for enumerating “LUN ID”.

[0094] FIG. 10 is a block diagram of memory system 1000c according to still another example of the disclosure. The memory system 1000c of FIG. 10 is similar to the memory system 1000 of FIG. 5 except that, the logic units 100 and 101 selectively operate in a blocking mode or a non-blocking mode of the sideband bus. Referring to FIG. 10, the logic units 100 and 101 have control ports BM0 and BM1, and host controller 300 has a corresponding control port BMH coupled to control ports BM0 and BM1, which transmits a control signal bm.

[0095] The control signal bm is used to control logic units 100 and 101 selectively operate in the blocking mode or the non-blocking mode. When the control signal bm has a high voltage level, logic units 100 and 101 operate in the blocking mode, and data signal da0 at input data port SI0 is delayed by a period and then passed to the output data port SO0. On the other hand, when the control signal bm has a low voltage level, logic units 100 and 101 operate in the non-blocking mode, and data signal da0 at input data port SI0 is directly passed to the output data port SO0.

[0096] FIGS. 11A and 11B are exemplary waveform diagrams of enable signal en, clock signal ck, control signal bm and data signals da0 and da1 of logic units 100 and 101 of FIG. 10. First, referring to FIG. 11A, the control signal bm has a high voltage level, hence logic units 100 and 101 operate in the blocking mode. At a rising edge r1 of clock signal ck, the content “A” carried by data signal da0 at input data port SI0 is latched by logic unit 100. Then, data signal da0 is delayed by a half of a clock cycle. At a falling edge f1 of clock signal ck, data signal da1 carrying the content “A”, is driven at output data port SO0 of logic unit 100. Data signal da1 is received at input data port SI1 of logic unit 101. In the blocking mode, data signal da1 lags data signal da0 by one clock cycle.

[0097] On the other hand, referring to FIG. 11B, the control signal bm has a low voltage level, hence logic units 100 and 101 operate in the non-blocking mode. The content “A” carried by data signal da0 is received at input data port SI0 by logic unit 100. Simultaneously, data signal da0 is directly passed to output data port SO0 of logic unit 100 to form data signal da1. Data signal da1 carrying content “A” is provided to input data port SI1 of logic unit 101. In the non-blocking mode, data signal da1 does not lag data signal da0.

[0098] FIGS. 12A and 12B are other exemplary waveform diagrams of enable signal en, clock signal ck, control signal bm and data signals da0-da2 of logic units 100 and 101 of FIG. 10, to illustrate status-reporting of logic units 100 and 101 in the non-blocking mode. Given that enumerating procedure for identifiers “LUN ID” of logic units 100 and 101 is completed, and logic units 100 and 101 have been assigned with final value of their identifier “LUN ID”. Such as, logic unit 100 is assigned with final value “0” of identifier “LUN ID”, and logic unit 101 is assigned with final value “1” of identifier “LUN ID”.

[0099] First, referring to FIG. 12A, during a period T1, final value “0” of identifier “LUN ID” of logic unit 100 is carried by data signals da0-da2, which are received at input data ports SI0, SI1 and MISO of logic units 100 and 101 and host controller 300. The final value “0” of identifier “LUN ID” indicates that, only the logic unit 100 with “LUN ID” of “0” is specified, while other logic units than logic unit 100 are not specified.

[0100] Then, during a period T2, data signals da0-da2 carry a command and are transmitted at input data ports SI0, SI1 and MISO. For example, the command is “Report Command” having binary bits of “00000101” which represents a hex number of “0x05”. The “Report Command” indicates that, a status of the logic unit 100, which is specified by “LUN ID” of “C”, will be reported to the host controller 300. In this example, only the status of the specified logic unit 100 is reported to the host controller 300.

[0101] Then, during a period T3, a status data SR0 which indicates the status of the specified logic unit 100, is provided in each of data signals da1-da2 in the non-blocking mode. The status data SR0

carried by the data signal da2 is received by the host controller **300** through the input data port MISO.

[0102] Then, referring to FIG. **12B**, during a period T4 which is subsequent to the period T3, final value “1” of identifier “LUN ID” of logic unit **101** is carried by data signals da0-da2, which are simultaneously received at input data ports SI0, SI1 and MISOI of logic units **100** and **101** and host controller **300**. Then, during a period T5, data signals da0-da2 carry a “Report Command” and are transmitted at input data ports SI0, SI1 and MISO. Then, during a period T6, output data port SO0 of logic unit **100** is tri-stated. Furthermore, a status data SR1 which indicates the status of logic unit **101** is provided in the data signal da2 in the non-blocking mode.

[0103] According to aforementioned various examples of the disclosure, identifiers “LUN ID” of logic units are enumerated utilizing sideband bus **200**, therefore port connections and wire bounding of logic units may be simplified, and memory system may achieve a lower cost. The identifier “LUN ID” may be dynamically enumerated and assigned during the power-on cycle or the reset cycle of logic units. The enumerating schemes for the identifier “LUN ID” utilizing the sideband bus **200**, may be applied to a memory unit **150** of a NAND flash type or a NOR flash type.

[0104] The sideband bus **200** may be executed with a protocol conformed to the UART or the USB, and the sideband bus **200** may have a higher operating frequency (e.g., up to 250 MHz), enumerating procedure for identifier “LUN ID” may be executed faster. Moreover, the status of each of logic units may be reported utilizing the sideband bus **200**, hence the main memory bus (i.e., the ONFI bus **250**) will not be occupied.

[0105] It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments. It is intended that the specification and examples be considered as exemplary only, with a true scope of the disclosure being indicated by the following claims and their equivalents.

Claims

1. A memory system, comprising: a plurality of logic units connected with one another in series to form a chain, each of the logic units has an identifier, an input data port and an output data port, and each of the logic units comprising: a control circuit, used to control a corresponding one of the logic units to receive a first data signal through the input data port, and transmit a second data signal through the output data port, and the output data port of the corresponding one of the logic units is coupled to the input data port of an adjacent one of the logic units, wherein each of the first data signal and the second data signal has a sequence of states, and the identifier of each of the logic units is enumerated according to a count number of a plurality of state transition edges of the states.
2. The memory system according to claim 1, wherein the chain has a type of Daisy Chain.
3. The memory system according to claim 1, wherein the output data port of the corresponding logic unit is coupled to a host controller to indicate that the corresponding logic unit is the last stage of the chain.
4. The memory system according to claim 3, wherein the host controller is configured to determine the total number of the logic units according to the state transition edges of the second data signal.
5. The memory system according to claim 3, further comprising: an aux controller, configured to provide the first data signal to the input data port of a first logic unit, and receive the second data signal from the output data port of a last logic unit to determine the total number of the logic units.
6. The memory system according to claim 5, wherein the logic units are accessed by the host controller through the aux controller.
7. The memory system according to claim 1, wherein the output data port of the corresponding logic unit is tied to a first voltage to indicate that the corresponding logic unit is the last stage of the

chain.

8. The memory system according to claim 1, wherein the input data port of the corresponding logic unit is tied to a first voltage to indicate that the corresponding logic unit is the first stage of the chain.

9. The memory system according to claim 1, wherein the input data port of the corresponding logic unit is unconnected externally and kept to a first voltage by an internal on-die weak connection to indicate that the corresponding logic unit is the first stage of the chain.

10. The memory system according to claim 1, wherein the input data port and the output data port of each of the logic units are operated with a sideband bus to enumerate the identifier, and the sideband bus is different from a main memory bus of the memory system.

11. The memory system according to claim 1, wherein the identifier is stored in a register and has a registered value, when a first state transition edge is detected, the registered value is an initial value, wherein each of the state transition edges is a rising edge or a falling edge.

12. The memory system according to claim 11, wherein when a subsequent state transition edge is detected, the registered value of the identifier is incremented.

13. The memory system according to claim 12, wherein when a last state transition edge is detected and a first period expires, the registered value of the identifier is determined as a final value according to the count number of the state transition edges, wherein the first period starts from the last state transition edge, and the first period is longer than two times of a duration of each of the states.

14. A memory system, comprising: a plurality of logic units connected with one another in series to form a chain, each of the logic units has an identifier, an input data port, an output data port and a clock port, and each of the logic units comprising: a control circuit, used to control a corresponding one of the logic units to receive a first data signal through the input data port and transmit a second data signal through the output data port, and receive a clock signal having a sequence of edges through the clock port, and the output data port of the corresponding one of the logic units is coupled to the input data port of an adjacent one of the logic units, wherein the corresponding one of the logic units transmits the second data signal carrying a first command, and the adjacent one of the logic units receives the second data signal, and the identifier of each of the logic units is enumerated according to the first command.

15. The memory system according to claim 14, wherein the chain has a type of Daisy Chain.

16. The memory system according to claim 14, wherein the input data port and the output data port of each of the logic units are operated with a sideband bus to enumerate the identifier, and the sideband bus is different from a main memory bus of the memory system.

17. The memory system according to claim 16, wherein the first command is an "Enumeration Command" of a SPI (Serial Peripheral Interface) protocol of the sideband bus, and the first command comprises a start bit and an end bit, wherein each of the edges of the clock signal is a rising edge or a falling edge.

18. The memory system according to claim 17, wherein the identifier is stored in a register and has a registered value, and the registered value is an initial value after a power-on cycle or reset.

19. The memory system according to claim 18, wherein when a first edge of the clock signal of the corresponding one of the logic units and the start bit of the first command are detected, the registered value of the identifier is kept unchanged, when an idle state which leading the start bit in the first data signal of the corresponding one of the logic units is detected at the first edge of the clock signal, the registered value of the identifier is incremented.

20. The memory system according to claim 19, wherein when the start bit to the end bit of the first command are verified to meet a content of the "Enumeration Command", the registered value of the identifier is determined as a final value, when the start bit to the end bit of the first command are verified not to meet a content of the "Enumeration Command", the registered value of the identifier is kept unchanged.

- 21.** The memory system according to claim 20, wherein the identifier of each of the logic units is enumerated after the power-on cycle or reset, and the final value of each of the logic units is kept unchanged until next power-on cycle or reset.
- 22.** The memory system according to claim 14, wherein the second data signal further carries a second command having a start bit and an end bit, and a status data of the corresponding one of the logic units is reported according to the start bit and the end bit of the second command and the clock signal.
- 23.** The memory system according to claim 22, wherein when the end bit of the second command is detected at one of the edges of the clock signal of the corresponding one of the logic units, the status data is transmitted through the second data signal.
- 24.** The memory system according to claim 23, further comprising: a host controller, coupled to a first logic unit and the last logic unit of the logic units, wherein when the logic units operate in a non-blocking mode, the status data of the corresponding one of the logic units is transmitted to the host controller.
- 25.** The memory system according to claim 24, wherein when the logic units operate in the non-blocking mode, one of the logic units is specified by the final value of the identifier, and the status data of the specified logic unit is carried in the second data signal and reported to the host controller.
- 26.** A memory system, comprising: a plurality of logic units connected with one another in series to form a chain, each of the logic units has an identifier, a clock port, a plurality of input data ports and a plurality of output data ports, wherein the output data ports of the corresponding one of the logic units are coupled to the input data ports of an adjacent one of the logic units; and a host controller, coupled to the input data ports of the first logic unit of the logic units, and coupled to the output data ports of the last logic unit.
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